

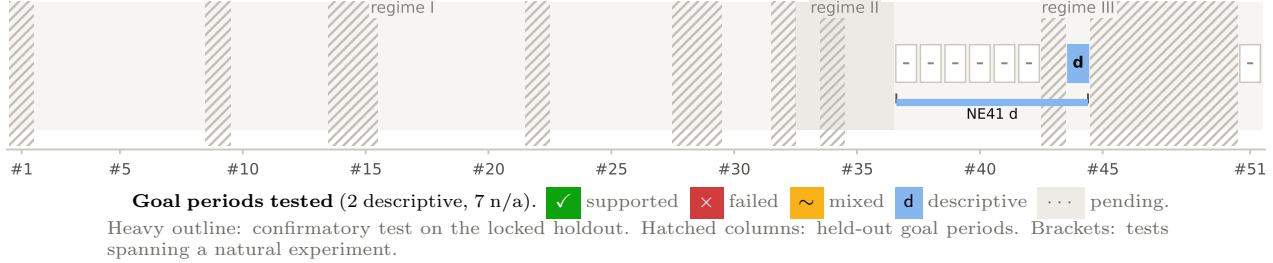
H139: A fast read kick on a slow private well predicts how content variance splits between the two

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

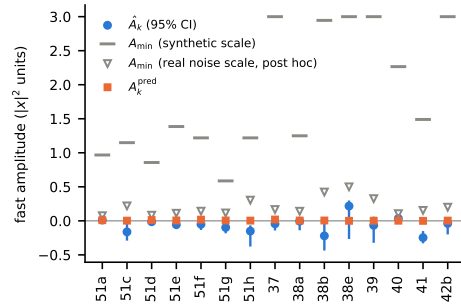
Bottom line. In 15 regime-III units the read-driven fast content variance predicted from H130's kick (median A_k^{pred} 0.0044, $\approx 2.5\%$ of the dynamic variance) lies 7 to $>1,000\times$ below the smallest resolvable fast amplitude, so HH382's variance-split test is untestable at this data size.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

H130 found that in #51 a read kicks the reader's content ($J_K = 0.044$) and the kick fades within ≈ 7 calls, while the agent's own well relaxes in ≈ 100 calls. HH382 asked whether that fast kick fixes, with no new fit, how much of an agent's content variance is fast. H139 measures the own content autocovariance at 1-call resolution in 15 regime-III units (#37–#42, #51) and compares its fast amplitude with the shot-noise prediction. A synthetic check on the real skeletons decided first whether the prediction is resolvable. It is not: the predicted fast amplitude is 7 to $>1,000$ times below the smallest resolvable one in every unit. The verdict is inconclusive (below resolution).



Observables. Fitted fast amplitude $\hat{A}_k$ (blue, 95% agent-day bootstrap CI) per testable unit, the predicted A_k^{pred} (orange) and the smallest resolvable amplitude $A_{\min}$ (gray bar: synthetic; open triangle: rescaled to the real noise level, post hoc). The prediction sits far below both resolution marks in every unit.

PHYSICAL MODEL

Two-rate Langevin relaxation (model 16) on the per-call clock. Content $x = s + k$: a slow part s held by the agent's private well (rate γ_s) and a fast part k made of read kicks of size J that decay at γ_k . For independent kick directions, Campbell's theorem fixes the fast amplitude from the read rate $\bar{r}$ (DQ1 ledger), J and γ_k (H130). The own autocovariance, corrected for the common drive, then has two exponentials and a plateau.

$$C(\tau) = A_k(1 - \gamma_k)^\tau + A_s(1 - \gamma_s)^\tau + B, \quad \tau \geq 1 \text{ call}$$

$$A_k^{\text{pred}} = \bar{r} J^2 / [1 - (1 - \gamma_k)^2], \quad Q_k = \hat{A}_k / A_k^{\text{pred}}$$

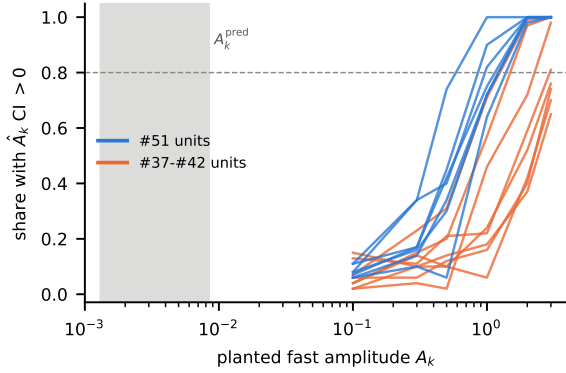
$$\text{S1: } A_{\min} > 2A_k^{\text{pred}} \Rightarrow \text{P1 untestable}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the read channel is not visible in an agent's overall content fluctuations: at about one read per call, H130's kick predicts a fast share of $\approx 2.5\%$ of the dynamic content variance (median), well inside the noise of $\sim 10^3$ – 10^4 statement pairs per unit. To see what reading does to an agent, measure the response along the sender's direction (H130's dose design), not the agent's total variance.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1 $A_{\min} > 2A_k^{\text{pred}}$	15/15 units; ratio 92->2,000 (6.8-9,993 real scale)	as predicted
S2 Q_k unbiased $\pm 30\%$	not resolvable at $\times 5, \times 20$	failed
S3 R_{fast} separates worlds	37-59% of replicates	failed
P1 $Q_k \in [0.5, 2]$ (kill)	$\hat{A}_k - 0.058 [-0.099, -0.017]$ vs 0.0044	untestable
P2 fast part $\propto$ reads	not computed	untestable
P3 slow share ≥ 0.8	13/15 units	non-diagnostic
P4 slow share is an agent constant	$\rho - 0.20, p 0.81$	failed
P5 two rates beat one	9/15 (null 34-58%)	non-diagnostic
N1 NE41 fast part erased	$R_{\text{slow}} 1.06 [0.98, 1.14]$	untestable



Second figure. Synthetic check on the real skeletons of 15 units (real statements, calls, ledger reads, resets; 50-100 replicates per world). Share of replicates in which $\hat{A}_k$'s 95% CI excludes 0, against the planted fast amplitude. Blue: #51 units; orange: #37-#42 units. The gray band is the range of A_k^{pred} . Detection reaches 80% only at $A_k \approx 0.6-3$.

GOAL PERIODS IN DETAIL

G44 d descriptive (no unit with ≥ 3 days).
 NE41 d descriptive. N1 is untestable (S1 fired in every unit...
 G37 - n/a (inconclusive: below resolution). S1 fired before...
 G38 - n/a (inconclusive: below resolution). S1 fired before...
 G39 - n/a (inconclusive: below resolution). S1 fired before...
 G40 - n/a (inconclusive: below resolution). S1 fired before...
 G41 - n/a (inconclusive: below resolution). S1 fired before...
 G42 - n/a (inconclusive: below resolution). S1 fired before...
 G51 - n/a (inconclusive: below resolution). S1 fired before...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The synthetic noise level was higher than the real one; rescaling to the real bootstrap SE (post hoc) still leaves $A_{\min}/A_k^{\text{pred}} \geq 6.8$. The drive correction subtracts conversational co-movement at short wall lags, and the lag-1 covariance sits below the lag-2 covariance in 14/15 units, so the constrained $\hat{A}_k$ is negative (post hoc). γ_{kick} was not re-estimated in the shared weeks. Agent-level slow shares are unstable.

ROUND 2

Measure the read echo along sender directions (H130's dose design), which has the power the variance split lacks. Explain the lag-1 dip (reply vs own-task alternation, or producing-call errors). Correct for the drive with unread partners only, and calibrate the synthetic noise to the real variance before new power claims.

- **Essence.** how much of an agent's moment-to-moment content is the echo of what it read.
- **H139-R1** Measure the echo where it is large.
- **H139-R2** The lag-1 dip.
- **H139-R3** Drive correction without conversation.
- **H139-R4** Calibrate the synthetic noise scale to the real $-x-^2$ and dynamic variance before any new power claim.

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